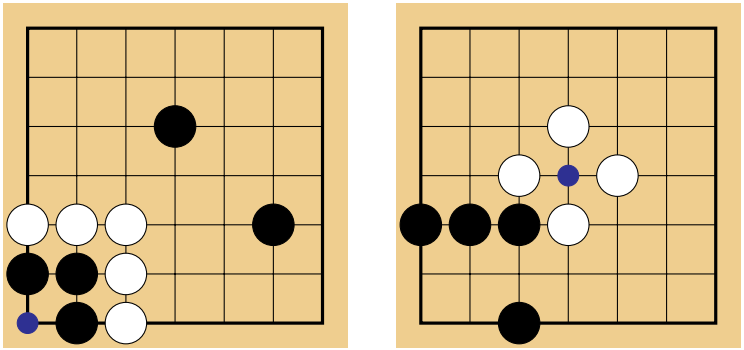


Figure 1.5 – In blue the illegal moves for Black as they are suicide.

There is, however, an exception to this rule: playing on an intersection with no liberties is legal if the move results in a capture. Upon removing the captured stones, the newly placed stone immediately gains liberties (see Figure 1.6). Furthermore, as seen in the figure, if both players were allowed to play on that intersection, the cycle of reciprocal captures could continue indefinitely. To prevent this loop, the *ko rule* is introduced. If Black were to play at coordinates $(4, 4)$ (blue dot in the figure), the board would revert to the exact position of the preceding turn.[1]

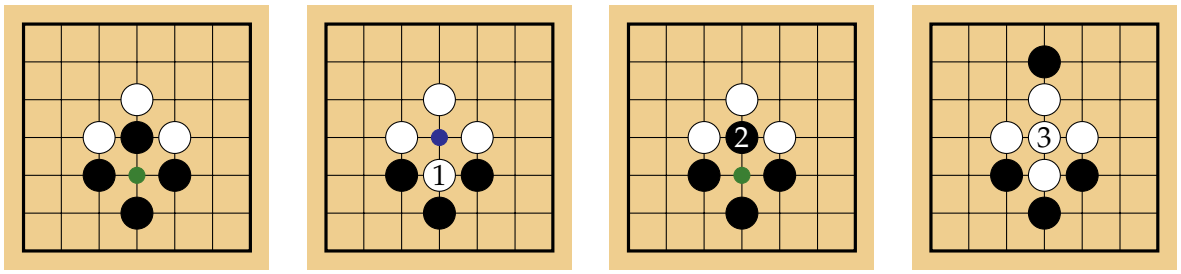


Figure 1.6 – An illegal position, as it reverts to a previous state. Following Black's move at $(4, 6)$, Black could recapture at $(4, 4)$ on a subsequent turn. White prevents this possibility by playing at $(4, 4)$.

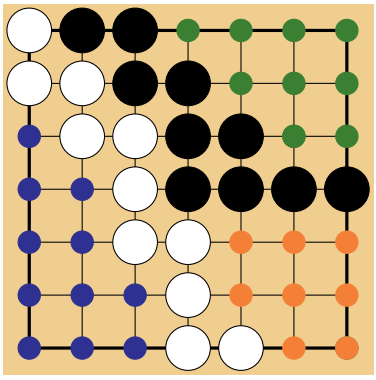


Figure 1.7 – Black territory, white territory and neutral.

To recapture, Black must first play elsewhere on the board (a ko threat). This generates a different board configuration; if the opponent responds, Black may then play at $(4, 4)$ to recapture, as the board state will no longer be identical to the earlier one (see the final board in Figure 1.6). An empty intersection (or a group thereof) is considered the *territory* of a given color if it is completely surrounded by stones of that color. Conversely, if an empty space is bounded by stones of both players, it is regarded as neutral, belonging to neither player (see Figure 1.7).

When two players of different skill levels face each other, a handicap is usually granted to the less experienced player. This advantage consists of playing with

[1] We will see that the standard *ko rule* proves insufficient in certain scenarios, as longer cycles can occur that repeat a previous board position without violating this basic rule (see Section 2.2.1).